Gender Pay Gaps in the Technology Industry: The Impact of Education Level and Experience

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Project Idea

Briefly explain your project, describe what you were trying to accomplish. Our project examines salary discrepancies between men and women in the technology industry, focusing on whether graduating from a top-tier university (e.g., Ivy League or top 20 schools) impacts gender pay gaps. With fewer women in STEM, they often face wage disparities and limited opportunities. We aim to determine whether elite education mitigates or exacerbates these inequalities by analyzing salary trends among Ivy League and non-Ivy League alumni.

Research Context & Objectives

This study explores whether attending a prestigious university reduces gender-based salary disparities in tech or if pay gaps persist regardless of educational background.

Goals

- Investigate gender-based salary discrepancies in the technology industry.
- Analyze whether attending a top 20 university influences salary differences between men and women in STEM.
- Examine the impact of factors such as education level, years of experience, and job title on salary disparities.
- Compare earnings among bachelor's degree holders in technology-related roles (e.g., Software Engineers, Data Analysts) across top 20 and non-top 20 university alumni.

Data

The dataset was created by Mohith Sai Ram Reddy, JG Sukumar, and Nikhileswar Sambangi. It is publicly available through Kaggle.

Dataset Overview * Rows (Observations): 6,704 * Columns (Predictors): 6 (Age, Gender, Education level[Bachelor's/Master's/PhD], Job Title, Years of Experience, and Salary)

Procedure

A. Data Preprocessing & Cleaning

Before analysis, we need to ensure the dataset is clean and structured properly.

1. **Handle Missing Values:**
    - Remove or impute missing values in key columns (**Age, Gender, Education Level, Job Title, Years of Experience, Salary**).
    - Since missing salary values can distort analysis, we may need to drop those rows.
2. **Filter for Relevant Data:**
    - Retain only **bachelor's degree holders** (removing master's and PhD graduates to focus on entry and mid-level roles).
    - Extract **technology-related occupations** (e.g., Software Engineers, Data Scientists, IT Analysts) by filtering the **Job Title** column.
3. **Convert Data Types & Standardization:**
    - Ensure **Gender** is categorized as **Male/Female** (if needed, map values consistently).
    - Convert **Years of Experience** and **Salary** to numerical data for statistical analysis.
    - Standardize salary data by adjusting for potential outliers using **z-score normalization or IQR filtering**.

B. Exploratory Data Analysis (EDA)

EDA helps us understand general trends, distributions, and potential disparities before deeper statistical modeling.

1. **Visualize Gender-Based Salary Distributions:**
    - Generate **box plots** and **histograms** of salaries for men and women to assess pay gaps.

- Compare mean and median salaries between genders.

2. **Analyze Salary by Experience & Gender:**
   - Create **scatter plots** and **regression lines** showing salary growth trends by years of experience for each gender.
   - Determine if women experience slower salary progression over time.

3. **Examine Education Level Impact:**
   - Compare salary distributions among bachelor's, master's, and PhD holders.
   - Analyze if women with higher education still earn less than men with equivalent degrees.

4. **Industry Roles & Pay Disparity:**
   - Identify the **most common job titles** for men and women.
   - Compare salaries across the same job titles to see if discrepancies exist within specific roles.

C. Statistical Analysis

To quantify the extent of gender-based salary differences, we will conduct statistical hypothesis testing and modeling.

1. **T-Test for Gender Salary Differences:**
   - Perform an **independent samples t-test** to compare male vs. female salaries.
   - Null Hypothesis (**H**): There is no significant difference between male and female salaries.
   - Alternative Hypothesis (**H**): Male and female salaries differ significantly.

2. **ANOVA for Education Level & Salary:**
   - Use **one-way ANOVA** to determine if there are statistically significant salary differences among **bachelor's, master's, and PhD holders**.

3. **Regression Analysis to Identify Salary Drivers:**
   - **Multiple Linear Regression Model:**
     - **Dependent Variable:** Salary
     - **Independent Variables:** Gender, Years of Experience, Education Level, Job Title
     - This will help quantify the impact of gender on salary while controlling for education and experience.
   - **Interaction Effect Analysis:**
     - Determine if the salary gap increases over time as experience grows.

D. Results Interpretation & Visualization

1. **Present Key Findings Using Charts & Graphs:**
   - Gender-based salary differences visualized with **bar charts, box plots, and line graphs**.
   - Regression results presented with **coefficient tables** and **statistical significance markers**.
2. **Highlight Any Alarming Trends:**
   - If a **statistically significant gender gap** exists, discuss how it manifests in different roles and experience levels.
   - Identify **occupations where the gap is most severe**.

E. Discussion & Conclusion

- **Summarize key takeaways:**
  - Do women earn significantly less than men in tech?
  - Does education level mitigate or worsen the gap?
  - How does experience affect salary progression for each gender?
- **Discuss limitations of the dataset** (e.g., missing university information).
- **Suggest future research directions** (e.g., incorporating company-specific or geographic data).

Roles

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